Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/21/2017
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Project Title:	Clinical Evaluation of the Newest Generation of a Non-Invasive ICP Meter (VITTAMED) in Subjects Undergoing Invasive ICP Measurement		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
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Space Biology Element:	None		
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Zip Code:	77030-3411	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Project aims/objectives: The overall aim of this study is to evaluate the accuracy of the newest generation of a non-invasive ICP (intracranial pressure) meter.
Specific aims: 1. Assess the accuracy of the newest generation of the non-invasive ICP meter (Vittamed) compared to simultaneous invasive ICP measurement in subjects with an Ommaya reservoir or undergoing lumbar CSF (cerebrospinal fluid) pressure measurements. 2. Assess the accuracy (sensitivity and specificity) of the non-invasive ICP meter to detect acute changes in ICP induced by body tilt in Ommaya reservoir undergoing invasive ICP measurements, or in patients undergoing lumbar CSF drainage.

Key findings:

Usability - The usability of the newest version of device is improved compared to the prototype version. In a previous

Task Description:	clinical validation study by Bershad et al. World Neurosurgery 2016 (https://), of 57 enrolled patients, only 24 (42%) had successful paired measurements due to various technical factors. These factors included poor headframe fit, inability to visualize two segments of the ophthalmic artery (intracranial and extracranial OA) in the same insonation angle, and software specific issues. In the current study, successful paired measurements in the similar patient population (lumbar puncture) was achieved in 17 of 17 (100%) of patients undergoing lumbar puncture and 4 of 4 (100%) of patients with Ommaya reservoir. Of 51 measurement attempts in the patients undergoing lumbar puncture, 30 (59%) were successful. The reasons for improved successful measurements are likely related to several factors including 1) Availability of different size headframes, 2) improved Doppler beam parameters. Based on the previous study by this PI, the company made it a priority to manufacture various size headframes based on known anatomical variations in skull morphology between individuals. Using a 3D printing method this was successfully implemented, and 3 different sizes of headframes were available to cover the full range of skull size. The improved Doppler beam allowed for a higher likelihood of finding an insonation angle whereby both an intracranial and extracranial segment of the ophthalmic artery were visualized together. Accuracy - The newer version of the device was less accurate compared to the previous version. The mean absolute difference between the vittamed ICP and the opening pressure (lumbar puncture) (n=17) was 9.35 mmHg SD (7.2 mmHg). The mean absolute difference between the vittamed ICP and the closing pressure (lumbar puncture) (n=12) was 5.1 mmHg (SD 5.1 mmHg). The vittamed had mean systematic bias of -9.1 mmHg as compared to the lumbar ICP at the opening pressure, and mean systematic bias of 0.48 mmHg at the closing pressure. Since the mean ICP of the invasive ICP was 24.6 mmHg at opening pressure versus 14.2 at the closing pressure, this indicates that the Vittamed gave falsely low ICP values when ICP is elevated compared to lower values. There are several possible reasons why the accuracy is less with this current version of the device compared to the previous version: 1. The methods for obtaining the Vittamed measurement changed in 2 ways that were designed to improve the usability of the method. First, in the previous study, a screening measurement was made prior to lumbar puncture in order to focus the subsequent paired measurement. The screening step used an external pressure range of 0 to 40 mmHg to cover a wide range of possible ICP. The value obtained from the screening measurement was then used to set the parameters to a narrower range that spanned 20 mmHg from the minimum to maximum external pressure. This may have improved the accuracy of the measurement, but at a cost of increased time to obtain an ICP measurement. Given the time limitations in astronauts and routine clinical practice, the PI used a different approach in which the screening step was omitted, and instead a wide range of external pressure was used during the paired measurement. Next, the redesign of the headframe may have had some unexpected influences on the relationship between the external pressure cuff, with the interface between the cuff and rigid headframe and/or cuff and orbital tissues. Fifth, the number of steps used 11 (0 to 36 by 4 mmHg) may have had an unexpected influence on the software algorithm calculation of ICP. Sixth, it appeared that a lower threshold for reporting ICP values was implemented by the manufacturer; thus, previously rejected measurements were included. Also, in some patients with highly elevated ICP, the external pressure range maximum was not high enough to cover a pressure beyond that of the invasive ICP, therefore it would not be possible to find a balance point in these patients. - The device was tested in 3 patients (4 total visits) with invasive ICP (Ommaya). The sample size was too small to draw any meaningful conclusions. However, the PI established a highly useful collaboration and platform for testing of future non-invasive brain physiology and ICP devices which will be useful for evaluating devices with possible spaceflight and Earth based applications. Impact of Key findings on hypotheses, objectives, and specific aims: The current version of the Vittamed ICP device is more usable than the previous version, but much less accurate. Further development is necessary before this device can be implemented for routine clinical use. Proposed research plan for the coming year: Preliminary indications suggest that Vittamed is developing new hardware and software in part due to this and other studies that will hopefully improve both the usability and accuracy of this method.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Vittamed device has significant potential for space and Earth based applications in that it can obtain an intracranial pressure measurement using non-invasive ultrasound. Further development is needed to improve the ease of use of the method, and accuracy of measurement. There are many populations on Earth that may benefit from a non-invasive ICP approach including idiopathic intracranial hypertension (pseudotumor cerebri), hydrocephalus, brain tumors, stroke, intracranial hemorrhage, and meningitis. It is also possible that this method could be implemented in developing countries using telemedicine assistance, or where neurosurgical expertise is either lacking or undesirable due to the risks.
Task Progress:	- IRB (Institutional Review Board) approval was successfully obtained at both clinical sites including Baylor College of Medicine outpatient clinic (lumbar puncture) and MD Anderson Cancer Center (MDACC) for patients with Ommaya reservoirs (Intracranial pressure). - Subjects were successfully enrolled at both sites. - Target enrollment was nearly achieved for lumbar punctures (17 of 20 target). - The target enrollment at MDACC was not achieved due to logistical issues that were encountered due to delays related to unanticipated administrative processes at MDACC, and slower than expected subject enrollment. - Preliminary data analysis completed; further analysis ongoing and will be presented at NASA Human Research Program Investigators' meeting 2018.
Bibliography Type:	Description: (Last Updated: 06/17/2026)

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